

CSC11006 - INTRODUCTION TO CLOUD COMPUTING SERVICES

PRACTICE

LAB1: GETTING STARTED WITH AWS BUDGET AND COST EXPLORER

I. General information

ID: LAB1
Period: 30 minutes
Deadline:

II. Outcome:

This lab will adapt to these following course's outcomes:

- G2.1
- G2.2

III. Describe:

A. Set up AWS Budget

AWS allows you to create several types of budgets: cost, usage, reservation instance or using template. In this section, we will practice creating each of them.

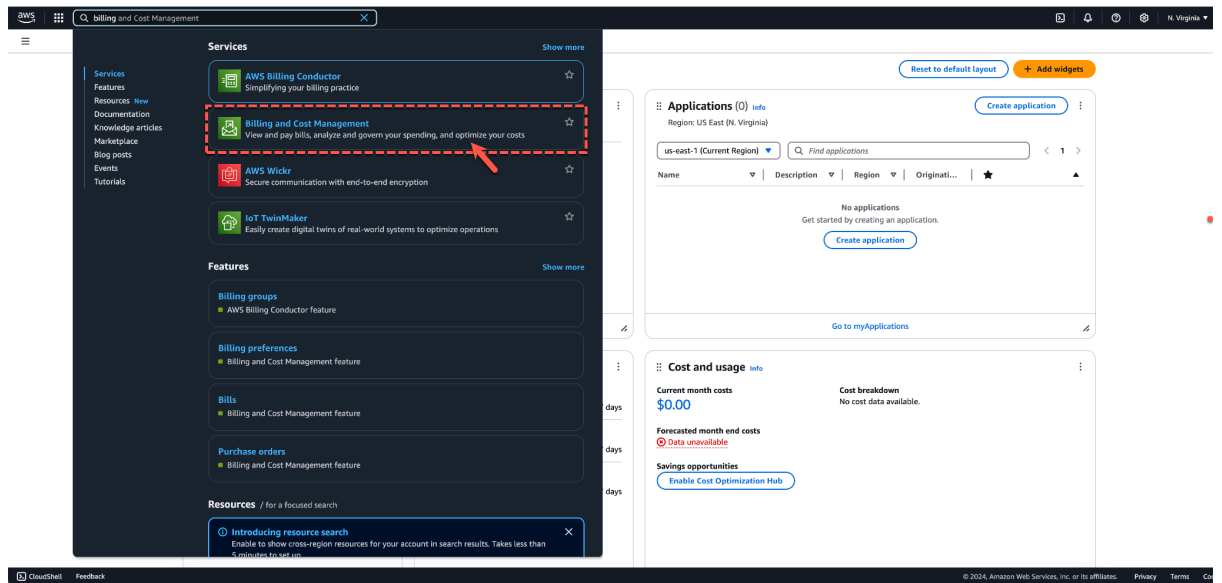
1. Create Budget by template:

You can create a budget using a template with pre-configured settings. Budget templates offer a streamlined approach to getting started with AWS Budgets, featuring a single-page workflow, in contrast to the more complex steps process required for customizing a budget (advanced).

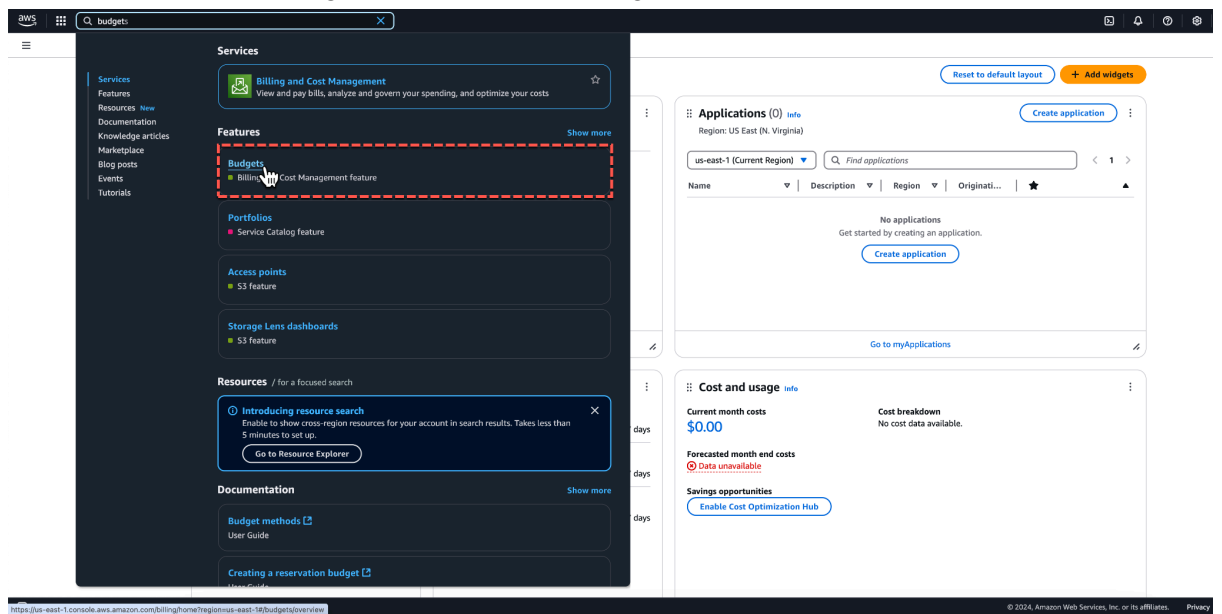
Step1. Sign in to the AWS Management Console and open the AWS Cost Management console at

<https://console.aws.amazon.com/cost-management/home>

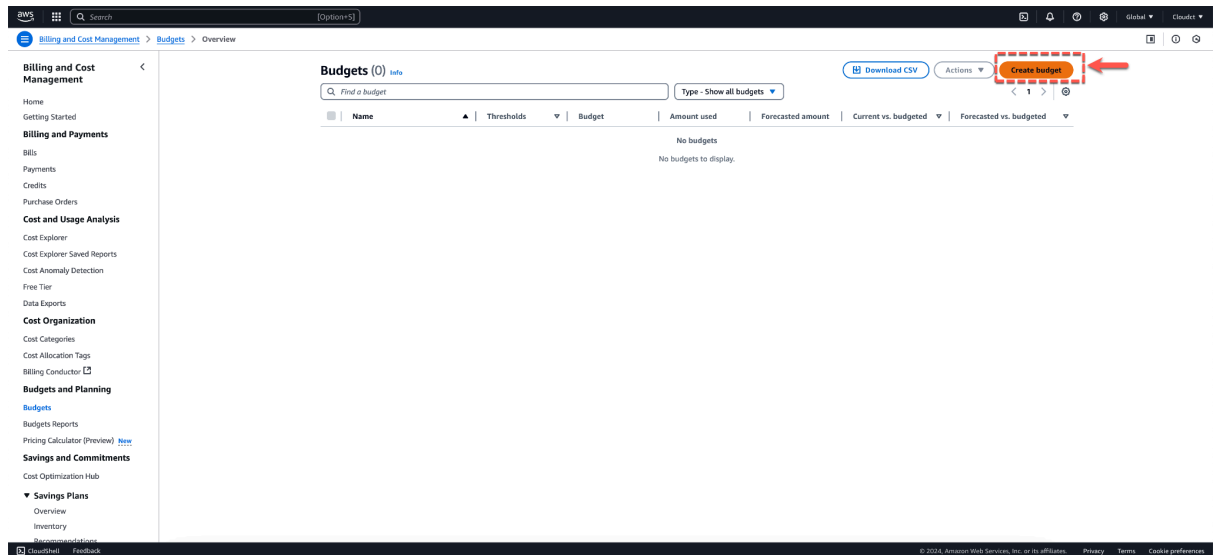
Locate and choose **Billing and Cost Management**



Step2. In the navigation pane, select Budgets.



Step3. At the top of the page, click Create budget.



Step4. Under Budget setup, choose Use a template (simplified).

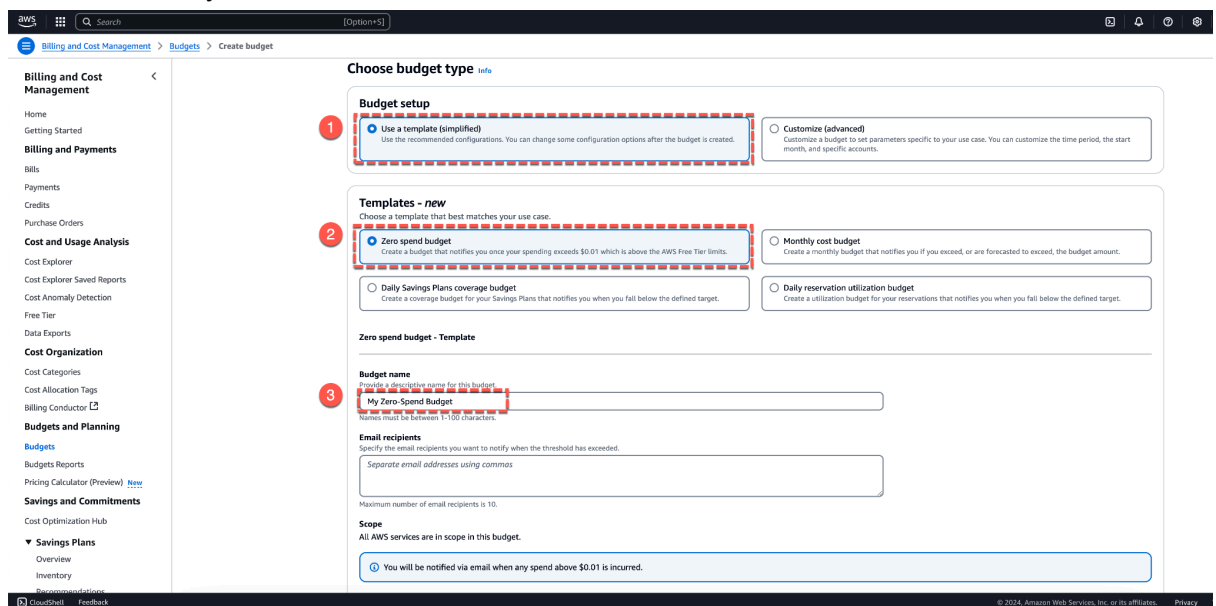
Step5. From the Templates section, select the template that best fits your needs:

Zero spend budget: Notifies you if your spending exceeds AWS Free Tier limits.

Monthly cost budget: Notifies you if your actual or forecasted spending exceeds your budget amount.

Daily Savings Plans coverage budget: Notifies you if your Savings Plans coverage falls below the defined target, helping you identify on-demand spending and consider new commitments.

Daily reservation utilization budget: Notifies you if your Reserved Instances utilization falls below the defined target, helping you spot unused hourly commitments.



Note: In this situation, I recommend you create a **Zero spend budget**.

Step6. Update the details and settings to match your specific needs.

Enter your email address, which you frequently check, to receive the alert when needed.
Step7. Click Create budget to finalize.

The screenshot shows the AWS Budgets 'Create budget' page. On the left is a navigation menu with categories like 'Billing and Cost Management', 'Cost and Usage Analysis', 'Cost Organization', and 'Budgets and Planning'. The main area shows the 'Zero spend budget' template selected. Fields include 'Budget name' (My Zero-Spend Budget), 'Email recipients' (youremail@gmail.com), and 'Scope' (All AWS services are in scope in this budget). A red dashed box highlights the 'Email recipients' field, and a red arrow points to the 'Create budget' button at the bottom right.

You will receive a message like this when your budget report has been created

The screenshot shows the AWS Budgets 'Budgets Reports' page. A green notification bar at the top states 'The budget report has been created successfully'. Below is a table with columns: Report name, Frequency, Budgets included, and Recipient(s). The table contains one row: 'My Zero-Spend Budget-report', 'Daily', '1', and '1'.

Report name	Frequency	Budgets included	Recipient(s)
My Zero-Spend Budget-report	Daily	1	1

2. Viewing budget:

You can quickly view the status of your budgets on the Budgets Overview page. Your budgets are displayed in a filterable table with the following information:

- Actual costs and usage for the budget during the current period
- Budgeted costs or usage for the period
- Forecasted costs or usage for the period
- A percentage showing your actual costs or usage relative to your budgeted amount
- A percentage showing your forecasted costs or usage relative to your budgeted amount

You can view detailed information in 2 ways:

2.1. First way:

Select a budget in the table to open a split-view panel with budget history and alert status on the right.

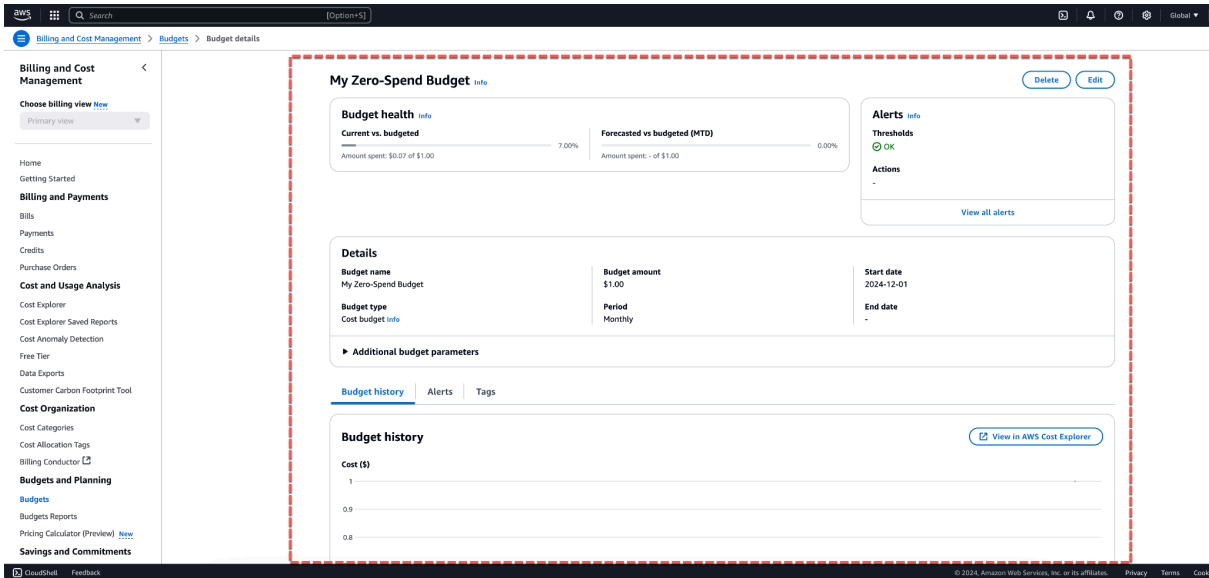
You can navigate between budgets using the buttons, but only one budget can be selected at a time. When multiple budgets are selected, the navigation buttons are hidden.

2.2 Second way:

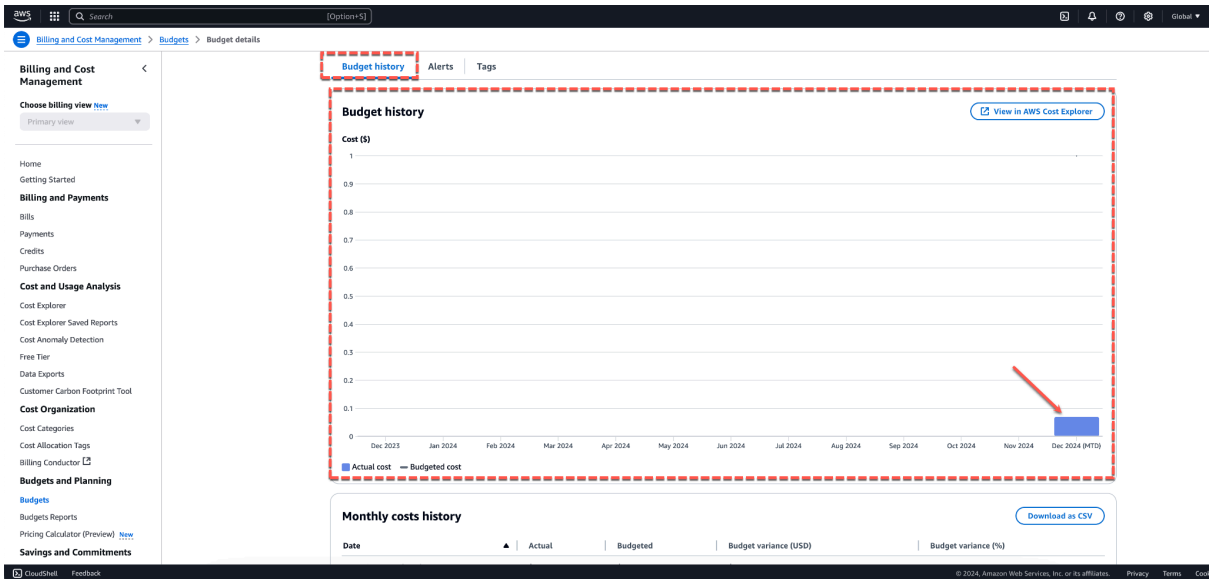
Choose your budget's name to see the budget details page. These information are included:

- **Current vs. Budgeted:** Your actual costs compared to your budgeted costs.

- **Forecasted vs. Budgeted:** Your forecasted costs compared to your budgeted costs.
- **Alerts:** Any notifications about your budget's status.
- **Details:** Information on amount, type, time period, and other parameters for your budget.



- **Budget History Tab:** A chart and table showing budget history—quarterly budgets show the last four quarters, and monthly budgets show the last 12 months. No history for annual budgets.



- **Budget Adjustments:** If you change the budgeted amount, the table reflects the latest budget.
- **Alerts Tab:** Details of any budget alerts, including conditions for exceeding the alert threshold.

Budget history **Alerts** Tags

Alerts (1) [Info](#) [View details](#) [Delete alert](#) [Edit alerts](#)

Review any exceeded thresholds and any pending actions below from your alerts.

Actual cost > \$0.01

Definition
When your actual cost is greater than **\$0.01 (1.00%)**, the alert threshold will be exceeded.

Threshold
⚠ **Exceeded**

Actions
-

Please notify if the threshold value is "Exceeded"

3. Deleting budget:

Step1. Login to AWS Management Console. Go to the AWS Budgets Console.

Step2. Select the Budget:

- In the Budgets dashboard, locate the budget you want to delete.
- You can use filters or the search bar to find a specific budget.

Step3. Open the Budget:

Click on the name of the budget to view its details.

Step4. Delete the Budget:

- Select the Delete button.
- Confirm the deletion when prompted.

The screenshot shows the AWS Budgets console interface. On the left is a navigation sidebar with categories like Billing and Cost Management, Cost and Usage Analysis, and Cost Organization. The main content area displays 'My Zero-Spend Budget' details, including budget health (Current vs. budgeted at 0.00%, Forecasted vs. budgeted (MTD) at 0.00%) and a 'Delete Budget' modal dialog. The dialog asks 'Are you sure you want to delete "My Zero-Spend Budget"?' and has 'Cancel' and 'Confirm' buttons. A red circle with the number '1' points to the 'Delete' button in the top right of the budget details panel, and a red circle with the number '2' points to the 'Confirm' button in the modal dialog. Below the modal, there is a 'Budget history' section with a table showing cost data.

B. Cost Explorer

AWS Cost Explorer is a powerful tool for analyzing and managing your AWS costs and usage.

1. Access Cost Explorer

- 1.1. Log in to the **AWS Management Console**.
- 1.2. Navigate to the **Billing and Cost Management Dashboard**.

1.3. Click on **Cost Explorer** in the left-hand menu.

2. Enable Cost Explorer (if not already enabled)

The first time you use it, you might need to enable it. Simply follow the prompts to activate it.

3. Explore Costs and Usage

3.1. Choose a View:

- AWS provides preconfigured reports, such as:
 - **Monthly costs by service**
 - **Daily usage**
 - **Forecasted costs**
- You can also create a **custom view**.

	Report name	Type	Time range	Time granularity	Grouped by	Filtered by
☐	Monthly costs by service	Cost and usage	Last 6 cal. months	Monthly	Service	-
☐	Monthly costs by linked account	Cost and usage	Last 6 cal. months	Monthly	Linked account	-
☐	Monthly EC2 running hours costs and usage	Cost and usage	Last 6 cal. months	Monthly	None	Usage type group: EC2: Running Hours;
☐	Daily costs	Cost and usage	Last 6 cal. months + Today to cal. month-end	Daily	None	-
☐	AWS Marketplace	Cost and usage	Last 12 cal. months	Monthly	Service	Billing entity: AWS Marketplace;
☐	RI Utilization	Reservation utilization	Last 3 cal. months	Daily	-	Service: Amazon Elastic Compute Cloud - Compute;
☐	RI Coverage	Reservation coverage	Last 3 cal. months	Daily	-	Service: Amazon Elastic Compute Cloud - Compute;
☐	Utilization report	Savings Plans utilization	Last 3 cal. months	Daily	-	-
☐	Coverage report	Savings Plans coverage	Last 3 cal. months	Daily	-	-

Select a report type

☒ **Cost and usage (recommended)**
The cost and usage report visualizes your aggregate costs across all AWS services. Use the filter dimensions to analyze all aspects of your AWS costs and usage.

Savings Plans reports

☐ **Savings Plans utilization**
The Savings Plans utilization report visualizes your aggregate Savings Plans utilization and allows you to set a custom Savings Plans utilization target. This report helps you understand how well you are using your resources.

☐ **Savings Plans coverage**
The Savings Plans coverage report visualizes your overall Savings Plans coverage and allows you to set a custom Savings Plans coverage target. This report helps you identify opportunities for savings.

Reservation reports

☐ **Reservation utilization**
The reservation utilization report visualizes your aggregate reservation utilization and allows you to set a custom reservation utilization target. This report helps you understand how well you are using your resources.

☐ **Reservation coverage**
The reservation coverage report visualizes your overall reservation coverage and allows you to set a custom reservation coverage target. This report helps you identify opportunities for savings.

Create Report

3.2. Set Time Range:

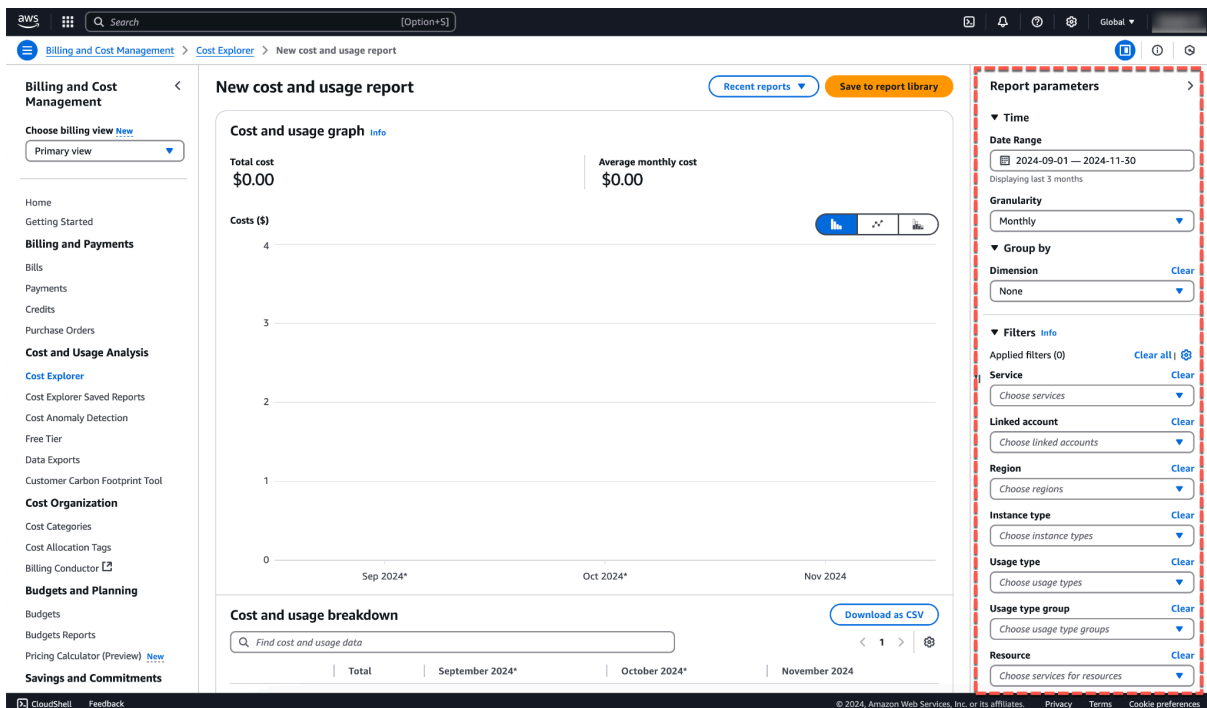
- Use the time range selector to focus on specific months, weeks, or days.

3.3. Filter Data:

- Filters let you narrow down your data. Common filters include:
 - **Services** (e.g., EC2, S3)
 - **Regions** (e.g., US-East-1)
 - **Linked Accounts** (in consolidated billing)
 - **Tags** (e.g., for project tracking)

3.4. Group Data:

- Group by categories like:
 - **Service**
 - **Usage Type**
 - **Linked Accounts**
 - **Tags**



4. Analyze Costs

- Use **visualizations** to identify trends and spikes in your spending.
- Drill down to specific services or time periods for detailed analysis.

5. Create Budgets and Alerts

- Combine Cost Explorer with AWS Budgets:
 - Set spending thresholds.
 - Receive email or SNS notifications if costs exceed these thresholds.

6. Delete Cost Saving Report

The screenshot shows the AWS Cost Explorer interface. On the left is a navigation menu with categories like 'Billing and Cost Management', 'Cost and Usage Analysis', 'Cost Organization', 'Budgets and Planning', and 'Savings and Commitments'. The main area is titled 'Reports' and shows a list of 10 reports. The report 'My_monthly_report' is selected, indicated by a blue checkmark in a box. A red circle highlights the 'Delete' button in the top right corner of the report list. Other reports include 'Monthly costs by service', 'Monthly costs by linked account', 'Monthly EC2 running hours costs and usage', 'Daily costs', 'AWS Marketplace', 'RI Utilization', 'RI Coverage', 'Utilization report', and 'Coverage report'.

7. Export Data

- Export your cost and usage data as CSV files for offline analysis.

8. Use Cost Anomaly Detection (Optional)

- AWS provides anomaly detection to alert you about unusual spending patterns.

The screenshot shows the AWS Cost Anomaly Detection 'Overview' page. The left navigation menu has 'Cost Anomaly Detection' highlighted with a red box. The main content area is titled 'Overview' and includes a section 'How Cost Anomaly Detection works' with three steps: 1. Create cost monitor, 2. View detection history, and 3. Analyze anomaly details. Below this is a 'Cost Anomaly Detection summary' section with four metrics: Anomalies detected (MTD): 0, Total cost impact (MTD): \$0, Total spend (MTD): \$0.08, and Total spend (vs. last month): →. The 'Detected anomalies' section shows 0 anomalies and a message: 'No anomalies. You don't have any anomalies at this time.'

Tips for Effective Use:

- **Tag your resources:** Use cost allocation tags to track spending by project, team, or environment.
- **Enable cost allocation reports:** Download detailed usage and cost data.
- **Monitor reserved instances and savings plans:** Identify opportunities for cost savings.